Psyc 210 Midterm Review

Lesson 1-6

Please check out the other resources that I have made to help you for the midterm.

I will cover the topics that were asked the most about in this presentation, and I will include practice questions

Links to other resources:

Psyc 210 Helpful Videos

PSYC 210 ULA Video Resources

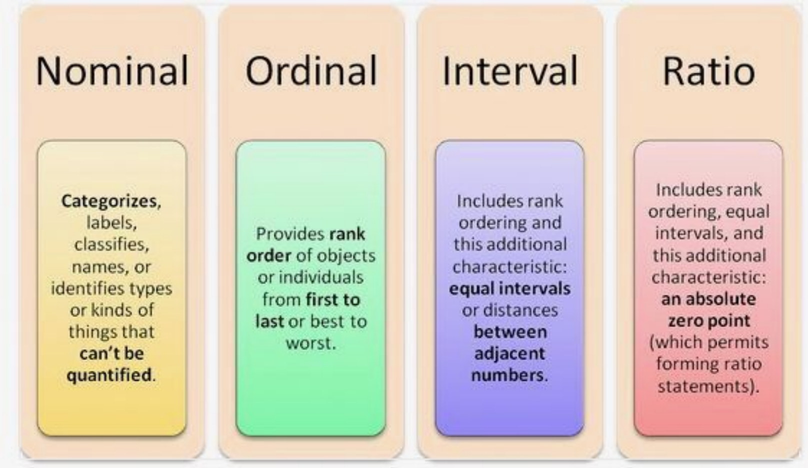
Psyc 210 Helpful Images

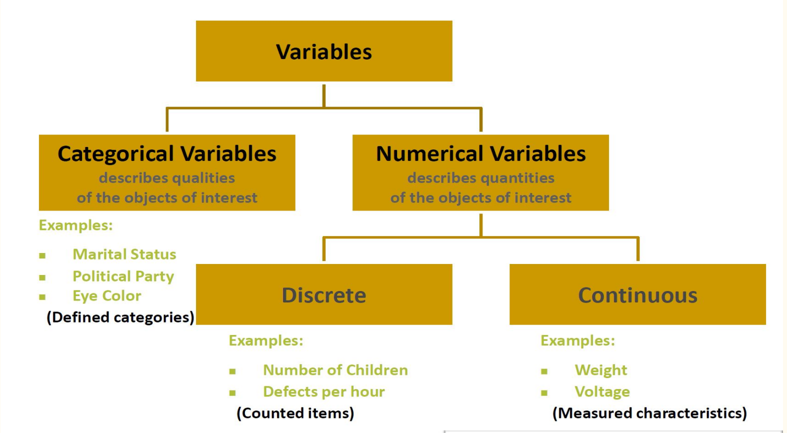
Intro Lesson

Types of Statistics

Know the difference between each type of statistic. Dr. Harrison explains these well in the slides.

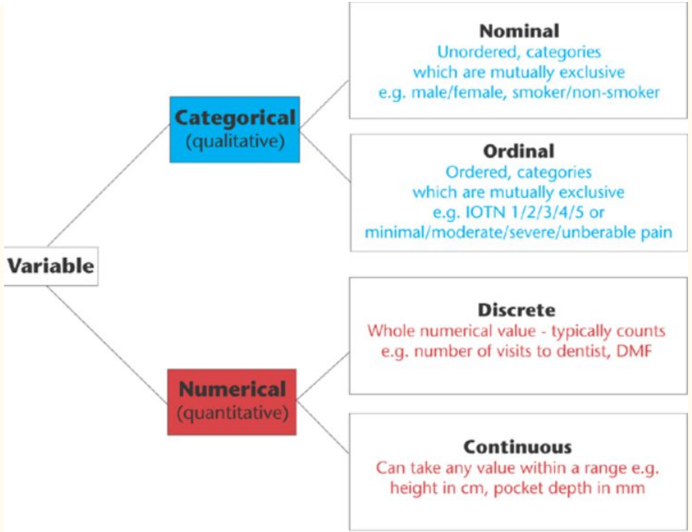
- Descriptive
  - **Describes** aspects of a sample
- Inferential
  - Draws **inferences** about the population
  - "Significance"





Levels of Measurement

Nominal	Ordinal	Interval	Ratio
"Eye color"	"Level of satisfaction"	"Temperature"	"Height"
Named	Named	Named	Named
	Natural order	Natural order	Natural order
		Equal interval between variables	Equal interval between variables
			Has a "true zero" value, thus ratio between values can be calculated



Lesson 1

Important Definitions

(Type of measurement and the best way to represent the distribution)

- Variable
  - Discrete vs Continuous
- Measurement
  - Nominal
    - Pie or Bar Graph
  - Ordinal
    - Bar graph
  - Interval
    - Histogram
  - Ratio
    - Histogram
- Data

Definitions

- Population Parameter
  - A number that describes something about an entire group or population
  - Mean or Standard Deviation
- Sample Statistic
  - set of individuals or objects collected or selected from a statistical population
- Sampling Distribution
  - the set of means from all the possible random samples of a specific size (n) selected from a specific population.

POPULATION	SAMPLE
The measurable quality is called a parameter.	The measurable quality is called a statistic.
The population is a complete set.	The sample is a subset of the population.
Reports are a true representation of opinion.	Reports have a margin of error and confidence interval.
It contains all members of a specified group.	It is a subset that represents the entire population.

QuestionPro

Parameter VS Statistic

When studying statistics, you might come across many terms that, if you aren't fully concentrated, will create a lot of confusion. For example, take parameter and statistic.

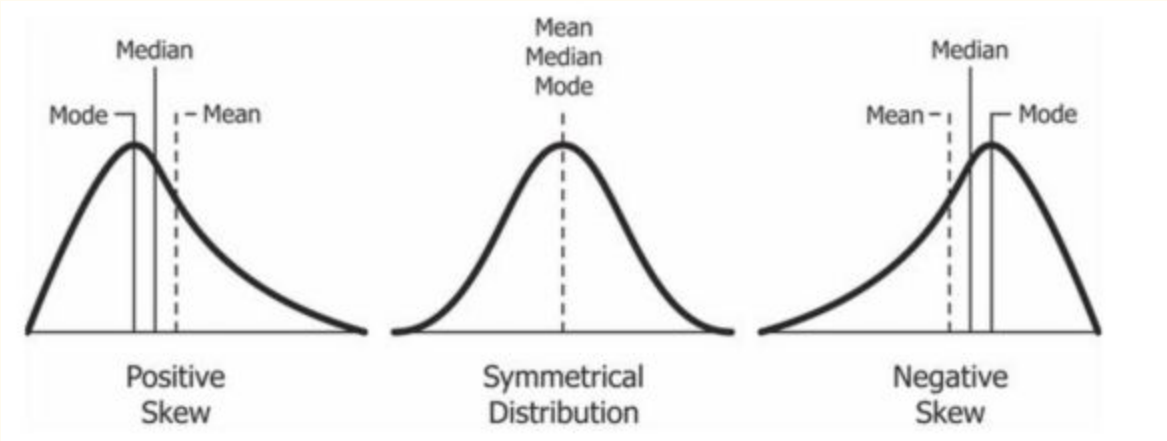
DEFINITION	DEFINITION
If you are talking about a PARAMETER, you are talking about the whole population.	A STATISTIC describes only a sample of the population.

How Distributions Differ

- Central Tendency
  - Mean
  - Median
  - Mode
- Variability
  - Range
  - Variance
  - Standard Deviation (SD)
- Skewness
  - Positive vs Negative
- Kurtosis
  - Mesokurtic
  - Leptokurtic
  - Platykurtic

Levels of Measurement & Central Tendency

Level of Measurement	**Central Tendency**
Nominal	Mode
Ordinal	Ordinal
Interval	Mean/SD
Ratio	Mean/SD



Term	Definition
Central tendency	The tendency for a set of values to gather around the middle of the set Generally measured by mean, median, and mode
Mean	Average $\sum x/n$ (sum of all values *[x]* over the number of values *[n]*) Should be applied to continuous data if normally distributed
Median	Middle value of an ordered sample of numerical values Extreme values do not affect the median as much as the mean, for example, length of stay, house prices Usually applied to numerical data (unless normally distributed)
Mode	Value that occurs most frequently Can be used for skewed numerical data or categorical data

Lesson 2

Important Takeaways

NEED TO KNOW!!!!
(By Doja Cat)

1. The mean can be deceptive. The mean only works with interval or ratio data

2. Means and medians only make sense with interval data

3. Know the difference between measures of central tendency and the three standard measures of variability

4. Statistics vs Population Parameters

    a. Statististics describes aspects of a sample

Lesson 3

Types of Distributions

- Population Distribution
  - The entire population
- Sample Distribution
  - Distribution of scores in a particular sample
- Sampling Distribution
  - Distribution of sample statistics

Z Scores

(Continued in Lesson 4)

- How a sample compares to the population (or sampling distribution)
- Use the standard normal distribution to estimate probabilities/percentiles

Lesson 4/5

Hypothesis Testing

Is my **sample mean** far enough away from my **population mean**...that I should assume the sample came from a different population?

Types of Hypotheses

- Null Hypothesis
  - Differences are due to chance
  - No real difference
- Alternative Hypothesis
  - Differences are not due to chance
  - There are REAL differences

Allows us to determine if a difference is due more to chance.

		Actual Situations	
		H0 Is True	H0 Is False
Our Decision	Reject H0	Type I Error a	Correct Decision!
	Fail to Reject H0	Correct Decision!	Type II Error b

Type I Error

- Always remains the same because we set it
  - Improving your measurement does not change alpha
  - Increasing your sample size does not change alpha
  - Alpha is always set at a particular value

Statistical Power

- The probability of rejecting the null hypothesis when it is in fact false
- Increase power by:
  - Reducing error variance
  - Increasing sample size

Degrees of Freedom

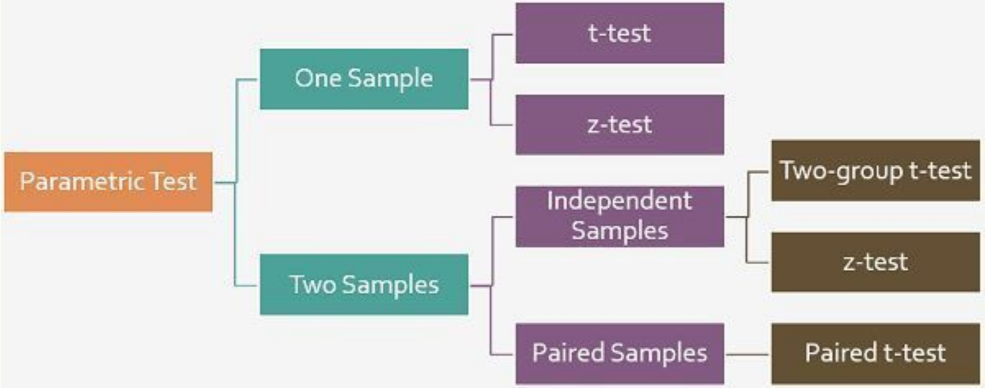
- Each data point in a sample is "free to vary"

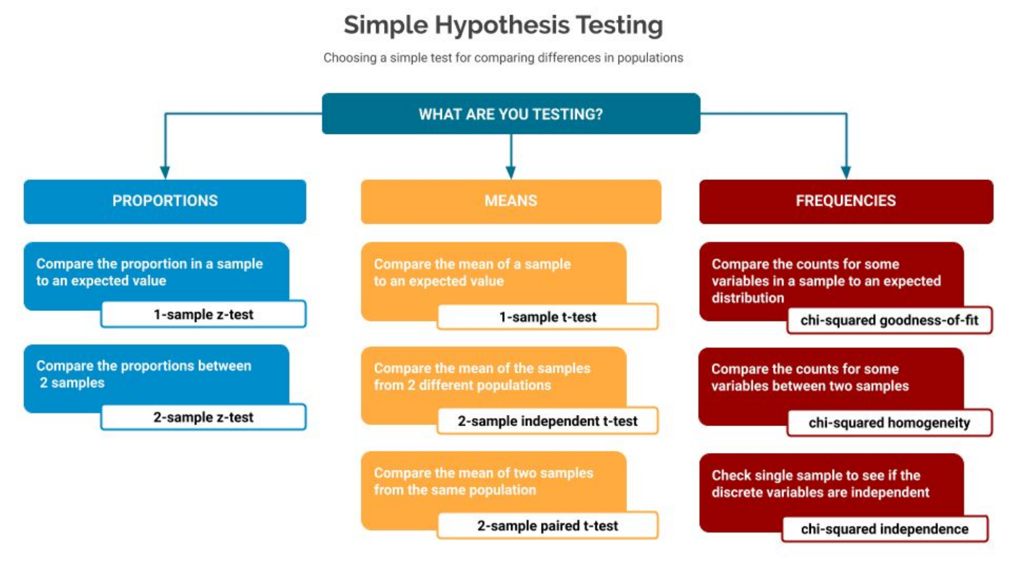
One-Tailed vs Two-Tailed Test

- A one-tailed test has the entire 5% of the alpha level in one tail (in either the left, or the right tail).
- A two-tailed test splits your alpha level in half
- A one-tailed test is appropriate if you only want to determine if there is a difference between groups in a specific direction.
- The main advantage of using a one-tailed test is that it has more statistical power than a two-tailed test at the same significance (alpha) level
- A two-tailed test is appropriate if you want to determine if there is any difference between the groups you are comparing.
- A two-tailed test uses both the positive and negative tails of the distribution. (tests for the possibility of positive or negative differences.)

Z vs T

- Use Z table if population SD is known
- Use T table if population SD is unknown


Parametric Test
One Sample
Two Samples
t-test
z-test
Independent Samples
Paired Samples
Two-group t-test
z-test
Paired t-test


Simple Hypothesis Testing
Choosing a simple test for comparing differences in populations
WHAT ARE YOU TESTING?
PROPORTIONS
MEANS
FREQUENCIES
Compare the proportion in a sample to an expected value
1-sample z-test
Compare the proportions between 2 samples
2-sample z-test
Compare the mean of a sample to an expected value
1-sample t-test
Compare the mean of the samples from 2 different populations
2-sample independent t-test
Compare the mean of two samples from the same population
2-sample paired t-test
Compare the counts for some variables in a sample to an expected distribution
chi-squared goodness-of-fit
Compare the counts for some variables between two samples
chi-squared homogeneity
Check single sample to see if the discrete variables are independent
chi-squared independence

Remember...

1. Must assume the mean difference is sampled from a normal distribution of mean differences
2. Assume the variances for each sample are the same (homogeneity of variance)
3. Must assume both samples were random and are **independent** from one another

What if the means are not independent?

- Correlated samples t-test
- No individual difference variance
- When to use a correlated samples t-test
    - Pre vs Post Test
    - Couples
    - Multiple Interventions
    - Related Samples
- Independent group t deals with both variance between groups and variance within groups - less power

Lesson 6

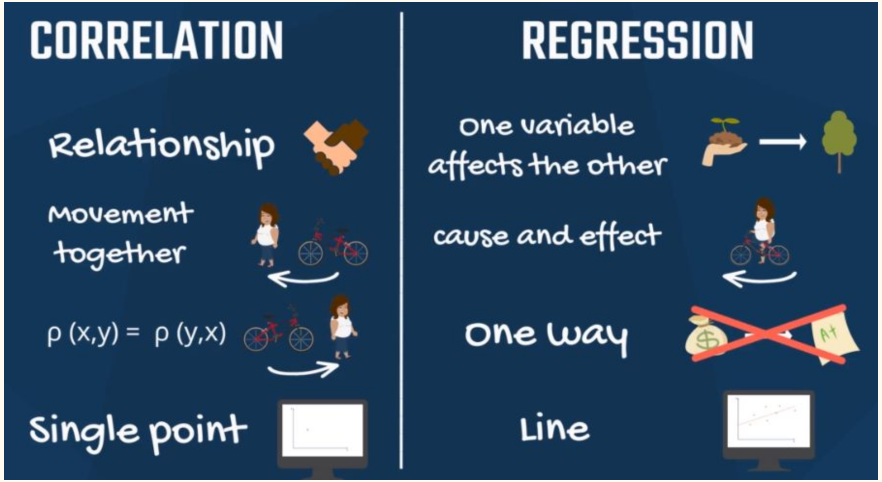
Correlation

- Statistical measure that expresses the extent to which two variables are linearly related

Limitations of Correlation

- Can be affected by the range of the X and Y variables
- Extreme scores can inflate correlations

	Correlation	Linear Regression
Purpose	Description, Inferential Statistics	Prediction, Designed Experiments
Statistic	r	r, R^2, R^2-adjusted
Variables	Paired	2, 3, or more
Variables	No differentiation between the Variables.	1 or more independent x's, 1 dependent: y = f(x)
Fits a line thru the data	Implicitly	Explicitly: y = a + bx
Cause & Effect	Does not address	Attempts to show



Basis for Comparison	Correlation	Regression
Meaning	Correlation is a statistical measure which determines co-relationship or association of two variables.	Regression describes how an independent variable is numerically related to the dependent variable.
Usage	To represent linear relationship between two variables.	To fit a best line and estimate one variable on the basis of another variable.
Dependent and Independent variables	No difference	Both variables are different.
Indicates	Correlation coefficient indicates the extent to which two variables move together.	Regression indicates the impact of a unit change in the known variable (x) on the estimated variable (y).
Objective	To find a numerical value expressing the relationship between variables.	To estimate values of random variable on the basis of the values of fixed variable.

learn.g2.com

Differences Between Correlation and Regression

Correlation	Regression
1 Relationship	1 One affects the other
2 Variables move together	2 Cause and effect
3 x and y can be interchanged	3 x and y cannot be interchanged
4 Data represented in single point	4 Data represented by line

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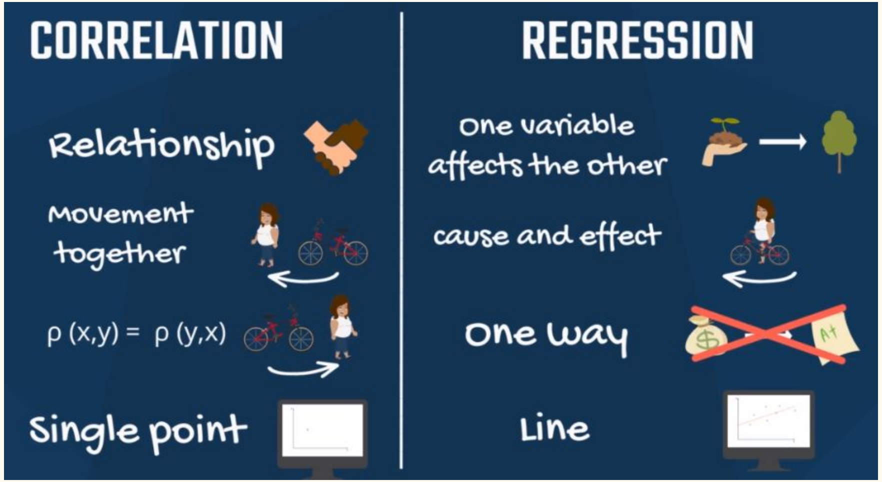

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Assumptions of Regression/Correlation

- Interval or Ratio scales
- Linear Relationship
- Normal distributions for X and Y
- Homoscedasticity

Correlation & Causation

- Correlation DOES NOT imply causation
- A third variable can come into play other than X and Y

Practice Problems

Problem 1

A researcher is interested in the eating behavior of rats and selects a group of 25 rats to be tested in a research study. The average score for the group of 25 rats is an example of a ______.

A. sample

B. statistic

C. population

D. parameter

Problem 2

The ______ is the preferred measure of central tendency when there are outliers or the data is heavily skewed.

A. Mode
B. Mean
C. Median
D. Range

Problem 3

Quantitative variables can be

A. Continuous
B. Discrete
C. Continuous and Discrete

Problem 4

Type of chocolate (white, milk or dark) is measured on a ____ scale; and the chocolate cocoa content in percentage is measured on a ____ scale.

A. Ordinal, Ratio
B. Nominal, Ratio
C. Nominal, Interval
D. Ratio, Interval

Problem 5

A correlation coefficient of -1.0 indicates...

A. A perfect positive relationship between the two variables
B. The correlation coefficient was incorrectly calculated
C. A perfect negative relationship between the two variables.

Problem 6

What inferential test is used to determine the significance of the relation between X and Y?

A. Z - test
B. T - test

Problem 7

What type of test is most appropriate when you want to compare an outcome between two groups with different people in each group?

A. Independent samples T - test
B. One Sample T - Test
C. Correlated Samples T Test

Problem 8

An assumption of the independent samples t-test is that the populations from which the samples are drawn have equal variability. This assumption is known as:

A. Power
B. Degrees of freedom
C. Homogeneity of variance
D. Correlation

Problem 9

If one were testing whether a particular sample was drawn from a population with a particular mean and the standard deviation of the entire population was known, one would use the _______test to statistically answer the question.

A. Z - test
B. T - test

Problem 10

Dr. Appleseed is interested in testing whether the proverb of “an apple a day keeps the doctor away” is true. She is specifically interested in adults 18 and in the United States. She recruits people from around her city to take part in her study.

1. What is the population?
2. What is the sample?
3. What is the variable?
4. What is the classification of the variable?
5. What scale of measurement should be used?

Answers

Answers

Numbers missed: 2,4

1. Statistic
2. Median
3. Continuous and Discrete
4. Nominal, Ratio
5. A perfect negative relationship between the two variables.
6. T - Test
7. Independent Samples T - test
8. Homogeneity of variance
9. Z - Test
10. American adults (18 and older), people recruited from Dr. Appleseed's city, number of apples eaten per day, continuous variable, ratio scale

Additional Practice Problems

And sources of where these problems are from

[Practice Test psyc 210 exam 1 Flashcards | Quizlet]

https://quizlet.com/348322858/psyc-210-final-exam-quiz-questions-flash-cards/

https://www.youtube.com/watch?v=o1WOhQwaIE0

I hope this presentation is helpful. Please let me know if there is anything else I can help you with.